

1. ResNet50 & EfficientNetV2-B0 (Our AI Brains)

A. ResNet50 (Residual Network - 50 Layers)

- * What it is: A famous 50-layer deep neural network invented by Microsoft Research.
- * The Problem it Solves: When neural networks get very deep (30+ layers), training signals disappear (Vanishing Gradient Problem).
- * The Solution (Residual Skip Connections): ResNet uses shortcut connections that pass data directly past layers $[F(x) + x]$. This allows signals to flow freely without fading away during training!
- * How We Used It in AgriVision AI:
 - Loaded ImageNet pre-trained weights.
 - Added Global Average Pooling (GAP) + Dense(256, ReLU) + Dropout(0.4) + Dense(38, Softmax).
 - Fine-tuned on 37,998 PlantVillage training images (saved in resnet_weights.npz).
- * PROFIT / ADVANTAGE: Heavyweight accuracy champion achieving 94.87% Top-1 Accuracy (Macro F1: 0.9335).

B. EfficientNetV2-B0 (Lightweight NAS Network)

- * What it is: A next-generation lightweight neural network designed by Google using Neural Architecture Search (NAS).
- * The Problem it Solves: ResNet50 has 25.6 Million parameters (large 96MB file size), which can be slow on low-cost smartphones.
- * The Solution (Compound Scaling): EfficientNet automatically balances network depth, width, and image resolution using search algorithms.
- * How We Used It in AgriVision AI:
 - Built & trained via src/models/train_efficientnet.py on PlantVillage split.
 - Saved fine-tuned weights to results/week2/efficientnetv2_weights.weights.h5 (28 MB).
- * PROFIT / ADVANTAGE:
 - Achieves 92.20% Top-1 Accuracy with only 5.9 Million parameters (77% smaller than ResNet50!).
 - Runs at 15.22 ms/img latency (2x faster inference for mobile field deployment).

2. Grad-CAM & Grad-CAM++ (Visual Focus Heatmaps)

A. Standard Grad-CAM (Gradient-Weighted Class Activation Mapping)

- * What it is: A visual explainability algorithm that generates glowing color heatmaps over an image.
- * The Problem it Solves: Neural networks are black boxes. Without Grad-CAM, you don't know if the AI looked at disease spots or background grass.
- * How It Works in AgriVision AI:
 - Extracts feature maps A^k from the final convolutional layer (conv5_block3_out in ResNet50).
 - Calculates 1st-order gradients of predicted class score y^c w.r.t feature maps.
 - Computes channel weights α_k and overlays a red/yellow heatmap (where red = highest AI visual focus).
- * PROFIT / ADVANTAGE: Provides visual proof to agronomists and farmers that the diagnosis is based on real leaf lesions.

B. Grad-CAM++ (Generalized Grad-CAM for Multi-Lesion Focus)

- * What it is: An advanced extension of Grad-CAM designed for multi-spot visual attribution.
- * The Problem it Solves: Standard Grad-CAM often highlights only ONE big central spot. If a leaf has 10 small scattered spots, standard Grad-CAM misses them!
- * How It Works in AgriVision AI:
 - Implemented in src/explainability/gradcam_pp.py using nested gradient tapes.
 - Calculates 2nd and 3rd order partial gradients w.r.t feature maps.
 - Weights positive first-order gradients using α_{ij} coefficients.
- * PROFIT / ADVANTAGE: Accurately highlights multiple non-contiguous lesion spots across the leaf blade with sharp boundary resolution (results/week3/xai_comparison.png).

3. Temperature Scaling & OOD Safety Guardrails

A. Temperature Scaling Calibration (T = 1.1959)

- * What it is: A post-hoc mathematical calibration algorithm (src/calibration/temperature_scaling.py).
- * The Problem it Solves: Raw neural networks are overconfident. They output 99.8% probability even when wrong!
- * How It Works in AgriVision AI:
 - Divides un-normalized output logits z_i by a scalar temperature $T > 0$ before Softmax: $p_{\hat{i}} = \text{Softmax}(z_i / T)$.
 - Fitted optimal $T = 1.1959$ on 8,161 validation samples by minimizing log-likelihood loss.
- * PROFIT / ADVANTAGE:
 - Drops Expected Calibration Error (ECE) from 1.09% down to 0.42% (61.61% relative calibration error drop!).
 - Model confidence scores now strictly reflect real-world diagnostic accuracy.

B. Uncertainty-Aware OOD Safety Guardrail (tau = 0.60)

- * What it is: A safety rejection filter that evaluates calibrated confidence scores.
- * The Problem it Solves: If a user uploads a photo of a hand, tractor, soil, or super blurry leaf, a raw model picks one of 38 plant classes.
- * How It Works in AgriVision AI:
 - Rejection Rule: If max calibrated confidence < 0.60 , the prediction is REJECTED as UNCERTAIN.
 - Tested on 30 non-leaf images (Tier 2) and 30 blurred leaves (Tier 3) in results/week1/ood_rejection_results.json.
- * PROFIT / ADVANTAGE: Displays a prominent warning banner and instructs the AI agronomist to issue an expert verification disclaimer.

4. Knowledge-Grounded RAG Engine & Full Stack App

A. Knowledge-Grounded RAG (Retrieval-Augmented Generation)

- * What it is: A framework that combines structured database retrieval with LLM generation (rag_engine.py).
- * The Problem it Solves: Direct LLMs (like ungrounded ChatGPT) hallucinate unverified chemical dosages, creating agricultural hazards.
- * How It Works in AgriVision AI:
 - Built a 38-class certified Agronomy Database (rag_knowledge_base.json) containing symptoms, chemical active ingredients, organic remedies, and prevention.
 - When a disease is diagnosed, retrieve_disease_context() fetches exact reference data and injects it into LLMs (Qwen2.5 / Mistral-7B).
- * ADVANTAGE:
 - Grounded RAG achieved 100.0% structured KB retrieval coverage across all 18 benchmark classes!
 - Direct LLMs without grounding risk fabricating active ingredient dosages (Strategy B human pass is PENDING).

B. Streamlit Web App, Voice Utilities & PDF Generator

- * Streamlit Dashboard (App.py):
 - Custom Glassmorphism CSS design (#0b1d16 gradient theme).
 - Sidebar radio buttons for dual model selection (ResNet50 vs EfficientNetV2) and dual XAI selection (Grad-CAM vs Grad-CAM++).
 - Batch processing gallery calculating field-level crop health scores.
- * Voice Utilities (voice_utils.py): Uses Web Speech API, OpenAI Whisper, and gTTS for Hindi & English voice interaction.
- * PDF Extension Generator (pdf_generator.py): Generates printable diagnostic health cards via FPDF2.
- * Verification Gate (verify_claims.py): Audits all 85 numeric claims in paper against JSON files (100% verified zero-fabrication science).